Assignment 2 : Git & GitHub Fundamentals

Group Report: Unit Converter

Group Size: 2 members
Chosen Theme: Unit Converter
Full Names: Kawtar Ziadi, Hibateallah Naboulssi
Student IDs: 162871, 171011
Course Codes & Sections: 3324 01-02
GitHub Usernames: Nandee44, hnaboulssi
GitHub Repository URL: https://github.com/hnaboulssi/group2practice2.git
GitHub Pages URL: https://hnaboulssi.github.io/group2practice2/

1. Project & Role Distribution

Our group chose the Unit Converter theme. The project converts temperatures between Celsius and Fahrenheit and distances between kilometers and miles. We followed the two-member assignment path.

- **Kawtar:** Responsible for temperature conversion: Celsius-to-Fahrenheit and Fahrenheit-to-Celsius functions and their unit tests.
- **Hibateallah:** Responsible for distance conversion: kilometer-to-mile and mile-to-kilometer functions and their unit tests.

Both members contributed through separate feature branches and pull requests targeting the main branch.

2. Branches, Commits & Contributions

Kawtar branch: kawtar-temperature-feature

implemented the temperature conversion functions and added tests for normal conversion, a boundary case, and invalid input. One commit for functions and another for tests with pytest

Hibateallah branch: hibateallah-feature

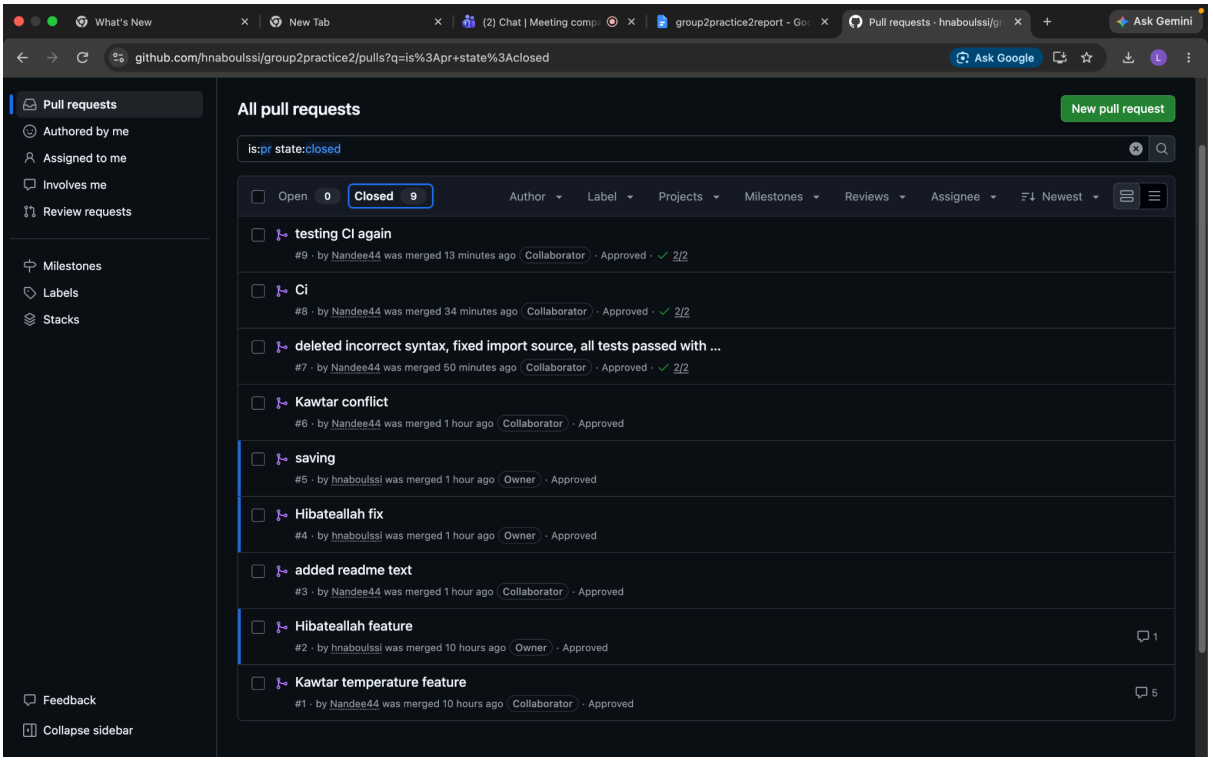
implemented the distance conversion functions and added tests for normal conversion, zero as a boundary case, One commit for functions and another for tests with pytest

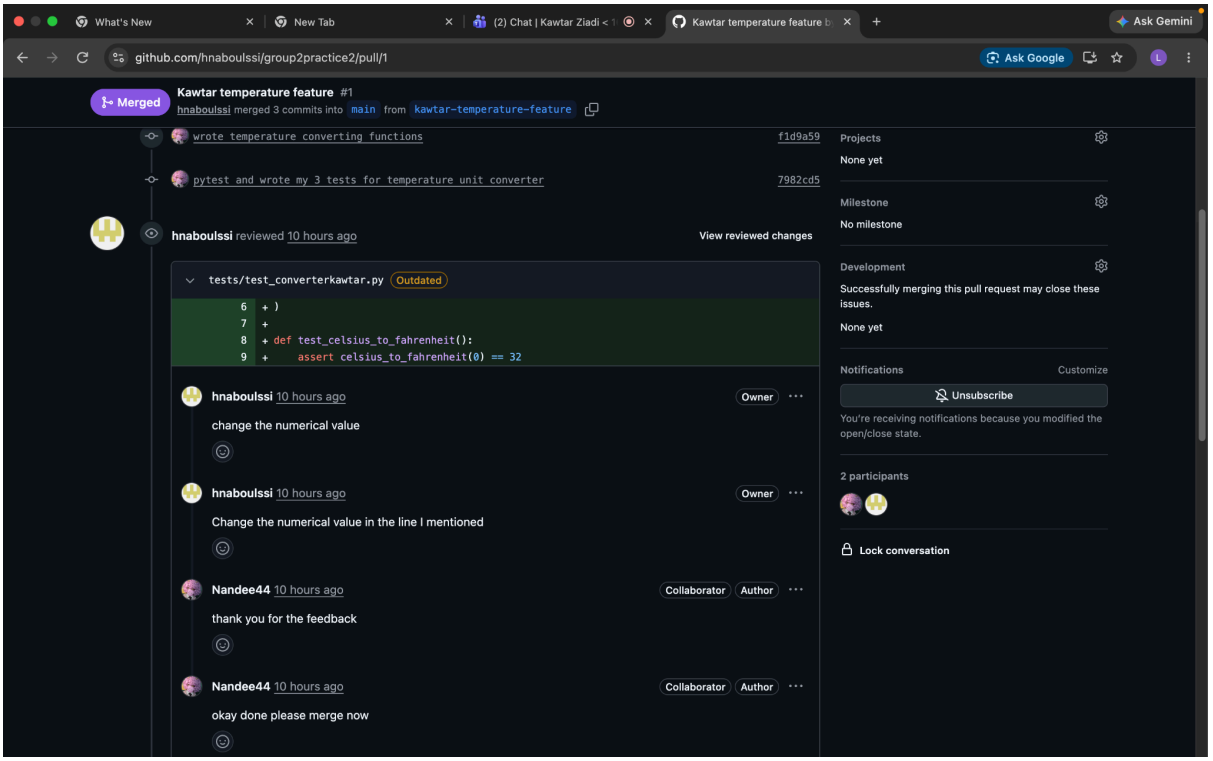
3. Pull Requests & Code Review

Each member submitted a pull request from their feature branch into `main`.

- **Kawtar's PR:** [PR number and title]. Reviewed by Hibateallah.
- **Hibateallah's PR:** [PR number and title]. Reviewed by Kawtar.

During the review process, the reviewer requested a change to the temperature conversion test. Kawtar updated the assertion from celsius_to_fahrenheit(0) == 32 to celsius_to_fahrenheit(100) == 212, committed the change, and pushed it to the same feature branch. This automatically updated the existing pull request. The reviewer then checked the revised code and merged.





4. Merge Conflict

To demonstrate a merge conflict, both feature branches changed the same README line:

Project status: Development in progress.

Each member changed the text in their own way. After the first branch was merged into main, the second branch fetched and merged the updated main branch, resulting in a conflict because both branches had modified the same line. After that the group resolved the merge conflict in the terminal.

```
intel@DESKTOP-1LHMLRP MINGW64 ~/group2practice2 (kawtar-conflict)
$ git fetch origin
remote: Enumerating objects: 20, done.
remote: Counting objects: 100% (20/20), done.
remote: Compressing objects: 100% (7/7), done.
remote: Total 12 (delta 5), reused 10 (delta 5), pack-reused 0 (from 0)
Unpacking objects: 100% (12/12), 2.75 KiB | 16.00 KiB/s, done.
From https://github.com/hnaboulssi/group2practice2
 * [new branch]      hibateallah-conflict -> origin/hibateallah-conflict
 * [new branch]      hibateallah-fix      -> origin/hibateallah-fix
   e445a81..c71a796  main                 -> origin/main

intel@DESKTOP-1LHMLRP MINGW64 ~/group2practice2 (kawtar-conflict)
$ git merge origin/main
Auto-merging README.md
CONFLICT (content): Merge conflict in README.md
Automatic merge failed; fix conflicts and then commit the result.

intel@DESKTOP-1LHMLRP MINGW64 ~/group2practice2 (kawtar-conflict|MERGING)
$ git merge origin/main
error: Merging is not possible because you have unmerged files.
hint: Fix them up in the work tree, and then use 'git add/rm <file>'
```

5. Unit Tests

We used pytest to test both feature areas.

Temperature conversion tests:

- Normal behavior: Checks that 100°C converts to 212°F.
- Boundary case: Checks that 32°F converts to 0°C.
- Invalid input: Checks that passing text instead of a number raises type error.

Distance conversion tests:

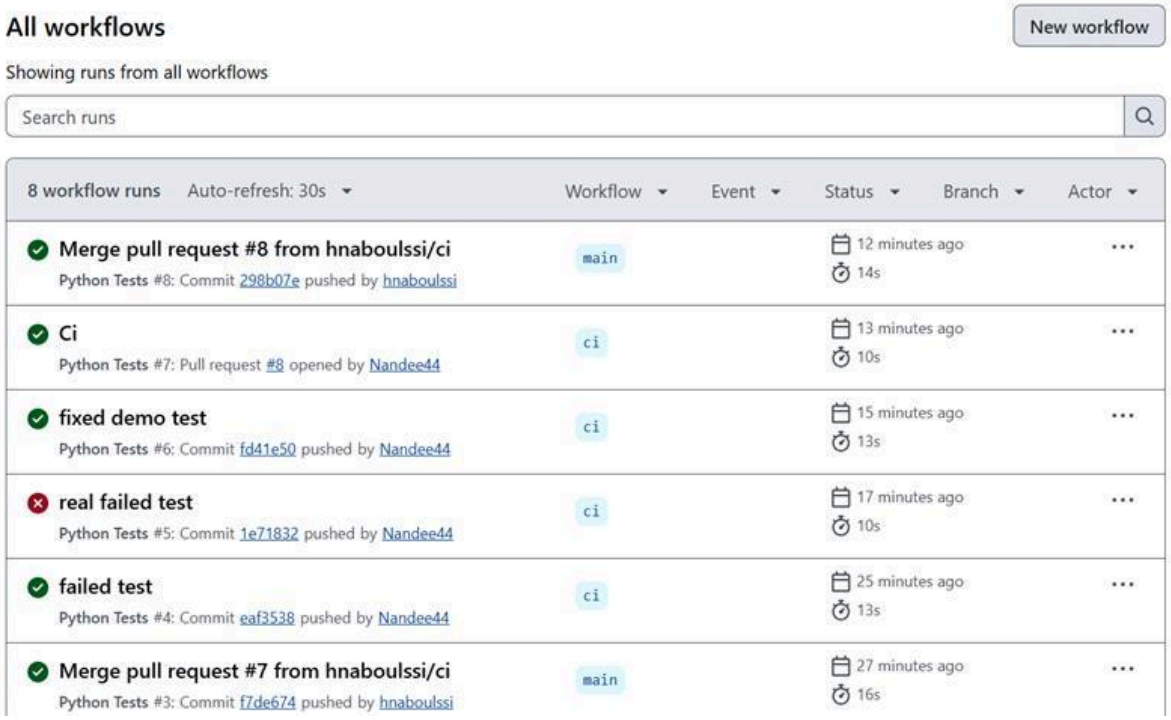
- Normal behavior: Checks that 10 kilometers converts to approximately 6.21371 miles.
- Boundary case: Checks that zero kilometers converts to zero miles.
- Invalid input: Checks that passing text instead of a number raises type error.

We ran the tests using python -m pytest or python3 -m pytest, depending on the local Python environment. In the beginning there were issues with testing, which were later fixed. Then a simulated failure in testing was made, then it was fixed as well.

6. Continuous Integration

The workflow uses pytest tests written in python and are automatically ran through CI.

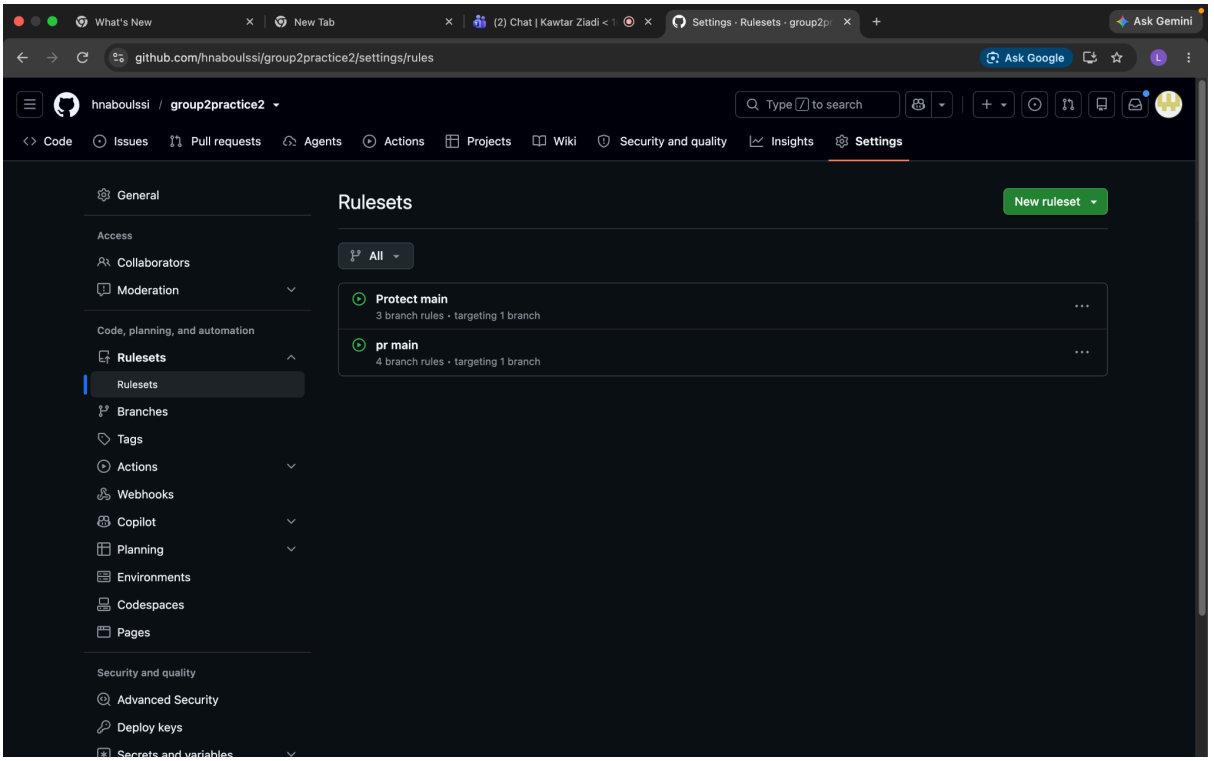
One CI run initially failed because of an intended test for failure. A later run completed successfully, confirming that the workflow could execute the test suite. After that, all tests are passed after each push, confirming that the project works.



7. Ruleset & Protected main

The repository's ruleset for main was configured to:

- Require a pull request before merging.
- Require at least one approval.
- Prevent deletion of the protected branch.
- Prevent force pushes.
- Require the GitHub Actions CI status check before merging



8. Continuous Deployment

We used GitHub Pages to publish the project's static website. The website presents the project title, team members, supported conversion features, and instructions. We made a final website change on a feature branch, submitted it through a pull request, and merged it into main. We then visited the live GitHub Pages URL and confirmed that the change appeared.

Live website: https://hnaboulssi.github.io/group2practice2/

9. Contribution Summary

Kawtar

- Feature: Temperature conversion.
- Branch: kawtar-temperature-feature.
- Tests: Normal conversion, freezing-point boundary, and invalid-input handling.

Hibateallah

- Feature: Distance conversion.
- Branch: hibateallah-feature.
- Tests: Normal conversion, zero boundary, and invalid-input handling.

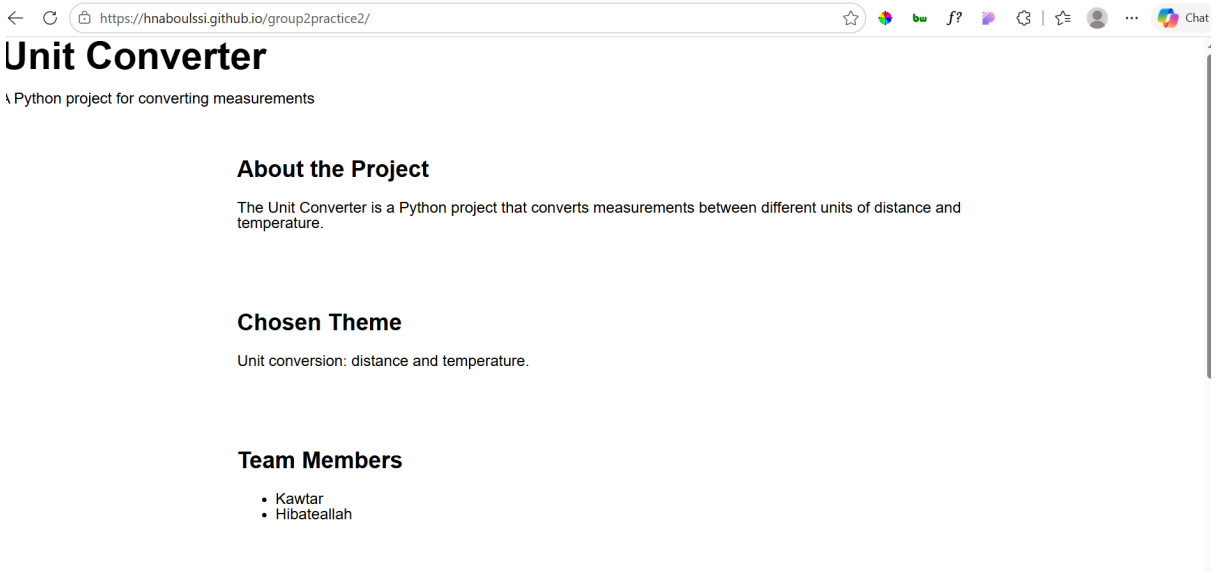
We reviewed each other’s PRs and Hibateallah requested changes before merging.

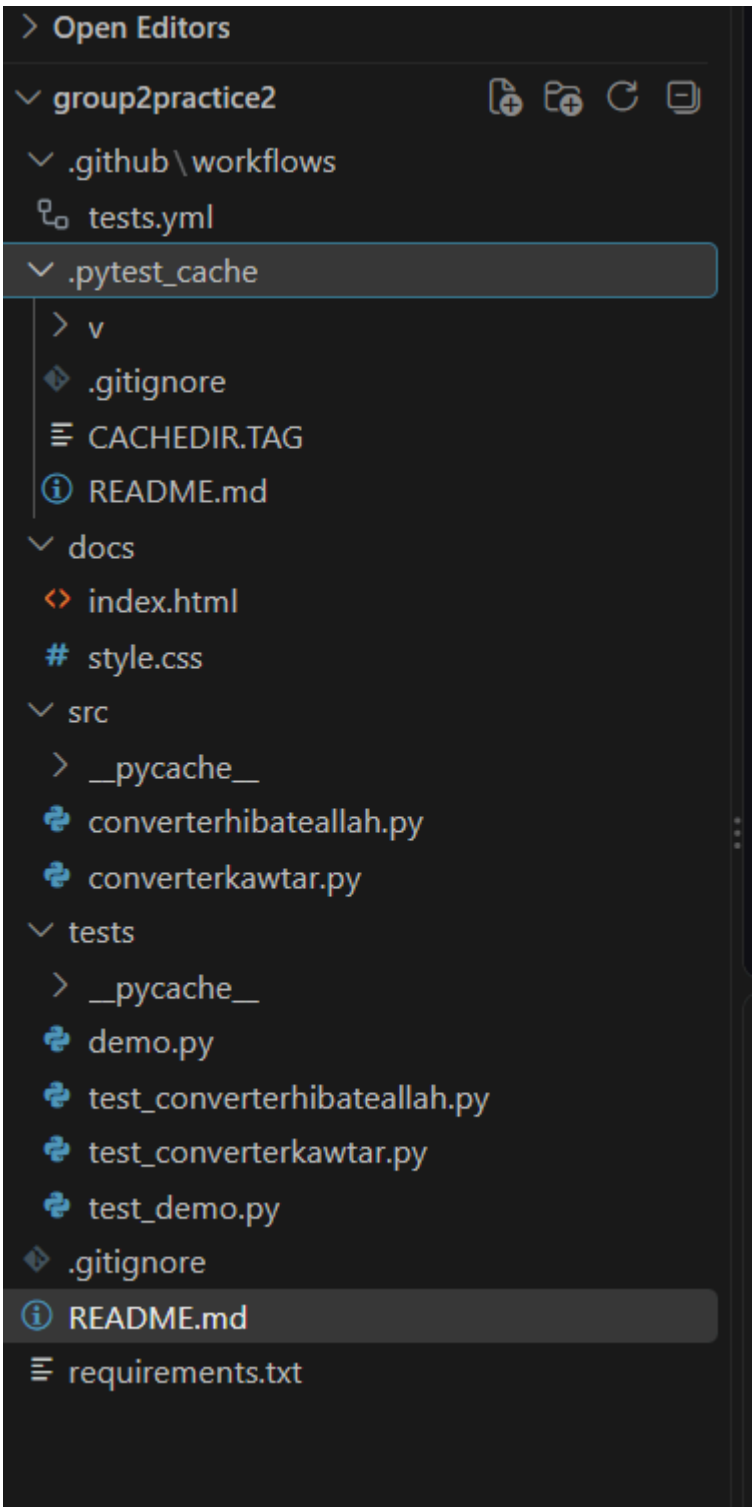
10. Fair Use of AI

We used AI to guide us through unexpected big problems like being unable to run pytest because we cloned the project too many times. We also used AI to help with commands we forgot and to verify that we are not messing up things. This is all a confusing and new process so this tool was very much needed.

11. Reflection

We were very challenged in multiple phases because this is all new to us both. We faced an issue with too much cloning which we fixed by deleting everything and starting over. We were struggling with the github UI for merging and everything but we dealt with it. Overall we learned a lot.




Open Editors
group2practice2
.github \ workflows
tests.yml
.pytest_cache
v
.gitignore
CACHEDIR.TAG
README.md
docs
index.html
style.css
src
__pycache__
converterhibateallah.py
converterkawtar.py
tests
__pycache__
demo.py
test_converterhibateallah.py
test_converterkawtar.py
test_demo.py
.gitignore
README.md
requirements.txt